

Fairness of Machine Learning Algorithms for the Black Community

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Abstract—This paper seeks to study the limits of the definitions of algorithmic fairness in relation to a protected variable, namely skin color. Discrimination towards the black community have existed for a long time. ML algorithms have only amplified or revealed existing discrimination. AI is a mirror that reflects the reality of our societies. The lack of a universal definition of algorithmic fairness makes it difficult to detect cases of discrimination in machine learning algorithms. We believe that independent or sensitive variables such as skin color are benchmarks that could be used to decide whether or not a decision is fair. We also recommend avoiding the use of proxy data.

Keywords—Algorithms; Black community; Ethics; Fairness.

I. INTRODUCTION

After the Emancipation Proclamation of 1862, slaves, the majority of whom were black, did not benefit from the same public services as the white community in the United States under the Jim Crow laws [1]. Before the Civil Rights Act of 1964 put an end to discrimination in hiring, phenomenon was prominent: some jobs were held only by whites and were inaccessible to blacks [2]. In 1964, the Illinois Fair Employment Practices Commission (FEPC) reviewer concluded that Motorola had discriminated against Leon Myart, a black American, in his application to work for Motorola as a “phase analyzer and phase shifter”. This is only one example, and the list is long.

Several researchers in the 1960s tried to find solutions to discrimination through fair testing [3]. A test is fair when the modification of the protected variable does not impact the test result. It must be noted that ML continues to perpetuate such discrimination.

Today, Machine Learning is increasingly used for decision making and has a significant impact on people's lives. It can disproportionately affect different groups or subgroups of the population, such as blacks. The question of the fairness of algorithms on the black community is of great interest. The issue of discrimination by AI is a very broad and deep subject. Algorithmic fairness is a critical study for under-represented minority groups, such as the LGBTQ community. However, our discussion will focus on the case of the black community.

This choice does not imply that discrimination on the black community is more frequent or that it is of greater importance than others.

The purpose of this paper is to raise awareness among the black community on the extent and difficulty, if not the impossibility, of having unbiased AI if strong measures are not undertaken.

First, we come back to the possible sources of bias of ML algorithms towards black people. Then, we follow by the role of data in the construction of equitable models and provide some examples of discrimination. Finally, we study the limits definitions of fairness for algorithms on variables such as skin color or race.

II. SOURCE OF BIAS IN MACHINE LEARNING

Bias comes mainly from three sources: Let f be an inappropriate predictive model, insufficient information in X to predict Y , or a set of information $(X; Y)$ related to the choice of the f -model. First, the presence of biases in the data may be the source of several biases in the outcomes of AI decisions. For example, African-Americans (Blacks) are subject to frequent and unwarranted contacts with the police, so the number of arrests and convictions of criminals may be biased by an under-representation of non-black people [4]. According to figures from the U.S. Department of Justice, one is twice as likely to be arrested if one is black than if one is white. A black person is five times more likely to get arrested without probable cause than a white person [5]. This is the case, for example, in facial recognition with photos of light and dark faces. A predictive model trained on biased datasets is highly likely to create disparities in police response.

Second, weighting, i.e. the choice of learning parameters and the selection of attributes. Algorithms can be biased by the values assigned to weights or parameters. For example, the PSA algorithm (the Public Safety Assessment) and the LSI-R algorithm (the Revised Inventory of Service Levels in Canada). The algorithms use criminal history and personal information such as age to determine an inmate's chances of recidivism. These algorithms do not include calculation [6]. Two inmates who have committed the same crime may have different results

based on data that is beyond their control, with no ability to challenge or evaluate the output of the algorithm.

The choice of learning parameters or the selection of attributes could be used to mask a belief about black people. Barocas and Selbst call this "masking" [7]. In this case, masking occurs through the deliberate selection of certain attributes for racial reasons. The protection of protected attributes may also discriminate against members of the class to be protected. For example, (Pujol et al, 2019) show that noise added for privacy may have a disproportionate impact on some groups compared to others if decisions are made using a different private version of the data from ϵ , under strict confidentiality constraints (smaller ϵ) [8].

Third, the choice of data and algorithms must meet the intended objective. Attribute selection should be based on objectives. It is therefore not prudent to use the same attributes for the construction of models with different objectives. The objective designates the result to be achieved through the use of AI. For example, in the United States, arrest data is often used to train crime prediction tools. This data is generated by police agencies. The majority of those arrested are black people. Unfortunately, an arrest does not necessarily lead to a conviction and does not imply that the person arrested is committing a crime [5]. In the above example the objective of the model is to predict crime. An arrest data alone is not appropriate for this purpose. In addition, the choice of objective should not be based on commercial or economic interests. For example, in the banking sector, the objective may be to maximize the number of loans repaid and not the creditworthiness of an applicant. The choice of objective could in this case discriminate against low-income applicants, who are in the majority of cases black.

Identifying bias is not easy. Already during the design of the model, bias detection is difficult. If the evaluation of the model's performance is based on its performance on validation data having the same source as the training data, the identification of biases would be difficult. Bias in the data must be assessed separately by evaluating the statistical properties of the data sets. Finally, the judgment of what is good or not depends on socio-cultural factors that are easily missed where there is a lack of diversity in AI teams. White people most often believe that their opinion is the right and fairest which is not necessarily true [9]. AI teams are predominantly white and male [10]. For example FAIR (Facebook AI Research) has noted an almost non-existence of black people [11] in its ranks. FAIR does not provide evidence of racial diversity in their teams. Diversity in AI teams is a way to reduce racial bias.

III. DATA

A thorough understanding of the origin of the data; its quality, pre-processing, processing and analysis, is fundamental to the construction of equitable predictive models for the black community. An example of the consequence of poor data quality can be found in the non-cautious use of Machine Learning by law enforcement coupled with the judicial system that reinforces historical race-based models [12]. In facial recognition systems, the quality of the image could also lead to disparities in predictions. The use of

historical training data tainted by racism may lead to racial discrimination [13]. Biased data reflects the reality of society. For example, in the USA, about one black man in three will be incarcerated during his lifetime [14]. The black community is often not sufficiently represented in the data used to train predictive models. This is the case of the data used by Static 99 [5]. Static 99 was trained with data from a sample in which only 12% were black people from the USA and 3% were black people living in Canada. The over-representation of blacks in some data causes serious problems and could lead to algorithmic bias. This is the case of the prison population in the US during the 2014 census [15]. Out of 318 million residents composed of 62.1% white, 13.2% black or African-American, and 17.4% Hispanic, the prison population was divided between 37% as black, 32% as white and 22% as Hispanic.

The consequences of the use of indirect data and social data are not to be neglected. The former could be detrimental to a black person because of the number of frequent police arrests of black people. The latter could discriminate against a black person applying for a high position due to the under-representation of black people executives on LinkedIn.

IV. DISCRIMINATION

A fair algorithm does not discriminate against a person on the basis of his or her belonging to protected groups. Belonging to a protected class can be based on racial origin. Racial discrimination generally involves an effort by one group to "produce self-esteem by lowering the status of another group" [16], inducing the emergence of privileges and benefits derived from belonging to the allegedly higher racial group. Statistical or direct discrimination explicitly takes into account protected attributes. AI developers may be influenced by their perception of black people. Some discriminatory actions are due to the fact that AI reflects the world as it is. The use of AI in the case of police or justice, without any measure of self-learning, reinforces historical patterns based on race [12]. Resource constraints can discriminate against Black people. For example, economically disadvantaged neighbourhoods are predominantly Black. In such a situation, such a decision should not only be perceived as discrimination against the community, but should also serve as a challenge to the Black community. A challenge in the sense that they must work intelligently and together to have economic influence and power like other communities, such as the Jewish community. In the absence of the use of protected characteristics, proxies (correlated variable) could be used to discriminate against black people. For example, companies recruiting students with the world's top MBAs is an example of discrimination via a proxy, when considering that very few black people have access to these programs. The following table provides a non-exhaustive list of the discrimination of black people in algorithms. In this table we have taken for our discussion four fields namely Police and Justice predictive, Hiring and publicity, Health, and Vision. In each field, at least two examples of algorithm discrimination are provided.

Table I. Example of discrimination

Fields	Reference
Police and Justice predictive	[17]; [18]; [19]
Hiring and publicity	[20]; [21]
Health	[22]; [23]; [24]
Vision	[25]; [26]; [27]; [28]

V. ALGORITHM FAIRNESS

The first article on algorithmic fairness to our knowledge dates back to 2008 with (Pedreshi et al. (2008)) [29]. Fairness was defined in a few ways, namely:

- Individual fairness: Giving similar predictions to similar individuals. Similarity predictions are defined by the distance between probability distributions on the outcomes generated by the predictive model. Therefore, the choice of the distance metric is of paramount importance and could compromise fairness. Individual fairness is very difficult to apply.

- Group and sub-group fairness: Group fairness is treating different groups equally. Group fairness allows each group to have the same probability of being chosen if we are in a recruitment case. Sub-group fairness aims to obtain the best properties of the group and the best notions of individual fairness. To know more about fairness, consult [30].

The three main families of algorithmic fairness definitions are anti-classification, classification parity, and calibration (see below). Beyond the families of algorithmic fairness, Aristotle makes a fundamental distinction between two criteria of fairness, namely horizontal fairness and vertical fairness [31]. Horizontal fairness is the principle that equal individuals should be treated equally, regardless of their ethnic origin or gender. In vertical fairness, unequal individuals are treated unequally (Corbett-Davies et al, 2018). We believe that the three fairness families suffer from a statistical limitation and find that instead of applying fairness rules, they argue that it would be preferable to treat at-risk individuals in the same way, based on the most statistically accurate estimates of risk that can be produced [32].

A. Anti-classification

Anti-classification does not explicitly take into account protected attributes such as race, gender, etc. A priori, anti-classification would seem to be the solution for the algorithmic fairness towards the black community because of the omission of the variable “race” or “skin color”. But anti-classification could be detrimental to the groups for which they were designed.

Even with the omission of the protected variable such as skin color or race, its effect will remain in the prediction via their correlation with the variables used. This is the case of Amazon with their 24-hour service advertisement that inadvertently discriminated against blacks because of their postal code. In the case of regression models, this is called “omitted variable bias”. Anti-classification could lead to discrimination based on taste.

B. Classification Parity

In classification parity, predictive performance is equal between the different groups. A risk assessment algorithm that predicts loan default could, for example, be required to produce similar false-negative rates for white and black applicants. Classification parity has limits to detect discriminatory algorithms. Moreover, the design of algorithms meeting this definition may have a negative impact on the well-being of communities. Classification parity is obtained by defining different decision thresholds for each group, and thus limits individual fairness.

C. Calibration and Statistical Parity

As for the calibration, the results of the risk estimation is independent of the protected attribute. Calibration does not guarantee a fair decision, although it is the right approach. Calibration provides a weak guarantee of fairness. It is possible to do a good calibration while discriminating individuals by misclassification. Statistical parity defines fairness as an estimate of the equal average probability of all members of protected groups. We believe that statistical parity ignores the individual characteristics and differences of protected group members.

In statistical parity the detention rates are equal for each racial group. This is a fundamental limitation of statistical parity because the detention rate should not be related to race at the risk of falling into positive discrimination.

VI. DISCUSSION

Risk assessment algorithms are often constructed from data that reflect human bias in past decisions and outcomes. It is therefore important to consider how problems with training data can corrupt risk scores. Beyond the biases caused by historical and proxy data, some algorithms incorporate biases in their codes through prejudices of AI team’s beliefs regarding the behavior of black people. Algorithms depend on their creators and their biases. This dependence can also make the algorithms unfair.

The lack of a universal definition of fairness shows the difficulty of solving the problem [33], the case of the COMPAS algorithm being a good example. The Propublica team using the ratio of false positives to true negatives between the group of black defendants and the group of white defendants came to the conclusion that the COMPAS algorithms were biased, while Flores and his colleagues demonstrated that the compass algorithm was fair because white and black defendants who received the same risk score had an equal probability of recidivism. Different cultural views and preferences on the notion of fairness make it more complicated to develop a single and universally acceptable definition of fairness for algorithms and their applications. As African wisdom says, equity is a contextual notion. For each application, the organization must define and contextualize what is fair or unfair. In the context of recruitment, an organization promoting cultural diversity would likely apply statistical parity. But if it advocates equal opportunity for all, the organization would be more likely to use anti-rankings or equity based on similarity (individual fairness). In any case,

fairness will remain a difficult notion to achieve in practice because AI models will never be entirely free of bias.

VII. CONCLUSION

From research on the fairness of tests to the fairness of machine learning algorithms, there is still no consensus on what is fair or unfair. Fairness of algorithms is obviously a difficult notion to have. In a world full of prejudice, algorithms will not be blind to race. In order for black people to be less discriminated, it is useful to eradicate a few prejudices. Black people have an important role to play. The majority of black people in the US are descendants of slaves. It is important that black people understand that the prejudices against them are not going to vanish as long as there are still prejudices about Africa. As said by African Wiseman, to know where we are going it is important to know where we are from.

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